



GOA COLLEGE OF ENGINEERING

"BHAUSAHEB BANDODKAR TECHNICAL EDUCATION COMPLEX"
FARMAGUDI, PONDA- GOA - INDIA

MATHEMATICS-III UNIT 1: MATRICES LECTURE 5

DEVELOPED BY MATHEMATICS FACULTY
DEPARTMENT OF SCIENCE & HUMANITIES
GOA COLLEGE OF ENGINEERING



MATRICES LECTURE 5

TOPICS TO BE COVERED:

- System of linear equations.
- Consistency of system of linear equations.
- System of homogeneous linear equations.
- Solution of system of homogeneous linear equations.
- Consistency of system of non-homogeneous linear equations.
- Solution of system of non-homogeneous linear equations.



System of Linear Equations.

A system of **m**-linear algebraic equations in **n**-unknowns $x_1, x_2, \dots, x_n$ is a set of equations of the form

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

.....

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

The numbers a_{ij} are known as coefficients and b_i are known as constants of the system.

If at least one b_i is non-zero then the above system is called system of non-homogeneous linear equations. If $b_i = 0$ for $i=1, 2, \dots, m$ then the system is said to be homogeneous linear of equations.



System of Homogeneous Linear Equations

$$\begin{aligned} i) \quad & 3x - 2y + 5z = 0 \\ & x + 2y + z = 0 \\ & 5x + 4y - z = 0 \end{aligned}$$

This is a homogeneous system of 3-linear equations in 3 unknown x , y and z .

$$\begin{aligned} ii) \quad & x_1 + 4x_2 + 5x_3 + x_4 = 0 \\ & x_1 - 2x_2 + 5x_3 - x_4 = 0 \\ & 2x_1 - 4x_2 + 3x_3 - x_4 = 0 \\ & x_1 + 4x_2 + 3x_3 - 2x_4 = 0 \end{aligned}$$

This is a homogeneous system of 4-linear equations in 4-unknowns. Here unknowns are x_1, x_2, x_3, x_4 .



System of Non-Homogeneous Linear Equations.

$$\begin{aligned} i) \quad & 3x + 5y + 3z = 5 \\ & x + y + z = 7 \\ & 5x - 2y - z = 1 \end{aligned}$$

This is a non-homogeneous system of 3-linear equations in 3 unknown x , y and z .

$$\begin{aligned} ii) \quad & x_1 + 4x_2 + 5x_3 + x_4 = 0 \\ & x_1 - x_2 + 5x_3 - 3x_4 = 0 \\ & 5x_1 + 4x_2 - 5x_3 - x_4 = 3 \\ & 6x_1 + 4x_2 + 5x_3 + x_4 = 2 \end{aligned}$$

This is a non-homogeneous system of 4-linear equations in 4-unknowns. Here unknowns are x_1, x_2, x_3, x_4 .



Matrix Representation of System

Let $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}$, $B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m \times 1}$ be two column vectors.

Let $A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}_{m \times n}$

Then the system

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

.....

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m$$



Matrix Representation of System

- Which can be written as

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \vdots & \vdots & \dots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ \vdots \\ b_m \end{bmatrix}$$

$$AX=B$$



Consistency of a System of Linear Equations

- A linear system of equations $AX=B$ is said to be consistent only if it has a solution.
- For a system of homogeneous linear equations $AX=0$
 $x_i = 0$, for $i=1,2\dots n$ is always a solution called zero or null solution. Therefore it is always consistent. This zero solution is called trivial solution. We are interested in a non-zero solution i.e at least one of the unknowns is non-zero.



Consistency of System of Homogeneous Linear Equations.

- The system of homogeneous linear equations $AX=0$ having m equations and n unknowns has a non-zero solution if and only if
Rank of $A = r < n$ (the number of unknowns).
- If consistent then the number of solutions are infinite.
- The infinitely many solutions are expressed in terms of $n-r$ variables.



Consistency of Homogeneous Linear System

Consider the system of homogeneous linear equations

$$3x+4y-z-6w=0$$

$$2x+3y+2z-3w=0$$

$$2x+y-14z-9w=0$$

$$x+3y+13z+3w=0$$

$$\begin{bmatrix} 3 & 4 & -1 & -6 \\ 2 & 3 & 2 & -3 \\ 2 & 1 & -14 & -9 \\ 1 & 3 & 13 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad AX=0$$



Consistency of Homogeneous Linear System

$$\bullet \begin{bmatrix} 3 & 4 & -1 & -6 \\ 2 & 3 & 2 & -3 \\ 2 & 1 & -14 & -9 \\ 1 & 3 & 13 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & -3 \\ 2 & 1 & -14 & -9 \\ 3 & 4 & -1 & -6 \end{bmatrix} \quad R_1 \leftrightarrow R_4$$
$$\bullet \begin{bmatrix} 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & -3 \\ 2 & 1 & -14 & -9 \\ 3 & 4 & -1 & -6 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 13 & 3 \\ 0 & -3 & -24 & -9 \\ 0 & -5 & -40 & -15 \\ 0 & -5 & -40 & -15 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - 2R_1, \\ R_3 \rightarrow R_3 - 2R_1, R_4 \rightarrow R_4 - 3R_1. \end{array}$$



Consistency of Homogeneous Linear System

$$\bullet \begin{bmatrix} 1 & 3 & 13 & 3 \\ 0 & -3 & -24 & -9 \\ 0 & -5 & -40 & -15 \\ 0 & -5 & -40 & -15 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 13 & 3 \\ 0 & -3 & -24 & -9 \\ 0 & -5 & -40 & -15 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_4 \rightarrow R_4 - R_3.$$

$$\bullet \begin{bmatrix} 1 & 3 & 13 & 3 \\ 0 & -3 & -24 & -9 \\ 0 & -5 & -40 & -15 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 13 & 3 \\ 0 & 1 & 8 & 3 \\ 0 & -5 & -40 & -15 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_2 \rightarrow -\frac{1}{3} R_2$$



Consistency of Homogeneous Linear System

$$\bullet \begin{bmatrix} 1 & 3 & 13 & 3 \\ 0 & 1 & 8 & 3 \\ 0 & -5 & -40 & -15 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 13 & 3 \\ 0 & 1 & 8 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} R_3 \rightarrow R_3 + 5R_2$$

- Rank of the matrix A= Number of non-zero rows in its Echelon form.
=2 < 4 (number of unknowns)
- Therefore system of equations is consistent i.e has a non-zero solution.
- The equivalent system of equations is
 - $x+3y+13z+3w=0$
 - $y+8z+3w=0$



Consistency of Homogeneous Linear System

$$x+3y+13z+3w=0$$

$$y+8z+3w=0 ; y=-8z-3w$$

Let $z=\alpha$ and $w=\beta$ then $y=-8\alpha-3\beta$ and

$$\begin{aligned} x &= -3y-13z-3w = -3(-8\alpha-3\beta)-13\alpha-3\beta \\ &= 24\alpha+9\beta-13\alpha-3\beta \\ &= 11\alpha+6\beta. \end{aligned}$$

The general solution is $x=11\alpha+6\beta$, $y=-8\alpha-3\beta$, $z=\alpha$ and $w=\beta$.

For any choice of α & β substitution in x , y , z and w gives a solution, which are infinite in number.

If $\alpha=1$ & $\beta=1$ then $x=17, y=-11, z=1$ and $w=1$ is solution.

If $\alpha=-1$ & $\beta=1$ then $x=-5, y=5, z=-1$ and $w=1$ is solution.



System of Non-Homogeneous Linear equations

The general system non-homogeneous linear equations is

$$a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2$$

.....

$$a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \text{ which can be}$$

written as

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \cdots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$



System of Non-Homogeneous Linear Equations

- The system can also be written as $AX=B$ where

$$A = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}_{m \times n}, X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{n \times 1}, B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}_{m \times 1}$$



System of Non-Homogeneous Linear Equations

Augmented matrix $[A|B]$ of system is obtained adding column matrix B to A

$$\text{i.e., } [A|B] = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & \dots & a_{2n} & b_2 \\ \dots & \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} & b_m \end{bmatrix}$$

Ex. $x_1 + 4x_2 + 5x_3 + x_4 = 0$

$$x_1 - x_2 + 5x_3 - 3x_4 = 0$$

$$5x_1 + 4x_2 - 5x_3 - x_4 = 3$$

$$6x_1 + 4x_2 + 5x_3 + x_4 = 2$$

For this system Augmented matrix is $[A|B] = \begin{bmatrix} 1 & 4 & 5 & 1 & | & 0 \\ 1 & -1 & 5 & -3 & | & 0 \\ 5 & 4 & -5 & -1 & | & 3 \\ 6 & 4 & 5 & 1 & | & 2 \end{bmatrix}$



Methods for Solving Non-homogeneous System of Linear Equations

Cramer's Rule:

This method can be used only when the number of equations and the number of unknowns are equal. Consider a system of n equations and n unknowns given by the matrix representation

$$\text{Let } \mathbf{AX} = \mathbf{B}$$

$$\text{Consider } \begin{vmatrix} b_1 & a_{12} & \dots & a_{1n} \\ b_2 & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ b_n & a_{n2} & \dots & a_{nn} \end{vmatrix} = \begin{vmatrix} a_{11}x_1 + \dots + a_{1n}x_n & a_{12} & \dots & a_{1n} \\ a_{21}x_1 + \dots + a_{2n}x_n & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1}x_1 + \dots + a_{nn}x_n & a_{n2} & \dots & a_{nn} \end{vmatrix}$$

By substituting $b_i = a_{i1}x_1 + \dots + a_{in}x_n$ $i=1,2,\dots,n$, in the first column.



Cramer's Rule

$$= x_1 \begin{vmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{vmatrix} + x_2 \begin{vmatrix} a_{12} & a_{12} & \dots & a_{1n} \\ a_{22} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{n2} & a_{n2} & \dots & a_{nn} \end{vmatrix} + \dots + x_n \begin{vmatrix} a_{1n} & a_{12} & \dots & a_{1n} \\ a_{2n} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{nn} & a_{n2} & \dots & a_{nn} \end{vmatrix}$$

$= x_1 D + x_2 \cdot 0 + \dots + x_n \cdot 0$ Since two columns are identical in the 2nd, 3rd n^{th} determinant.

There for $D_1 = x_1 D$ thus $x_1 = \frac{D_1}{D}$ where D is the determinant of square matrix A and D_1 is the determinant of square matrix obtained from A by replacing first column by column matrix B.

Similarly it can be proved that $D_i = x_i D$. For $i=1, 2, \dots, n$.



Cramer's Rule

Thus $x_i = \frac{D_i}{D}$ where D is the determinant of square matrix A and D_i is the determinant of square matrix obtained from A by replacing i^{th} column by column matrix B .

Thus if the determinant D of the square matrix A is non-zero i.e A is non-singular, then the solution of the system of non-homogeneous linear equations $AX=B$ is $x_i = \frac{D_i}{D}$ for $i=1,2,3...n$. Where D is the determinant of A and D_i is the determinant of a square matrix obtained from A by replacing the i^{th} column by column matrix B .

Note: This method involves computing determinants and for large systems consumes huge resources and time.



Example

- Solve by Cramer's Rule

$$x + y + z = 11$$

$$2x - 6y - z = 0$$

$$3x + 4y + 2z = 0$$

Solution: Consider $AX=B$

Where,

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -6 & -1 \\ 3 & 4 & 2 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 11 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{Let } D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & -6 & -1 \\ 3 & 4 & 2 \end{vmatrix} = 11 \neq 0$$

Therefore the system has Unique solution.



Example

$$\text{Now } D_1 = \begin{vmatrix} 11 & 1 & 1 \\ 0 & -6 & -1 \\ 0 & 4 & 2 \end{vmatrix} = -88$$

$$D_2 = \begin{vmatrix} 1 & 11 & 1 \\ 2 & 0 & -1 \\ 3 & 0 & 2 \end{vmatrix} = -77$$

$$D_3 = \begin{vmatrix} 1 & 1 & 11 \\ 2 & -6 & 0 \\ 3 & 4 & 0 \end{vmatrix} = 286$$

Then solution is

$$x = \frac{D_1}{D} = \frac{-88}{11} = -8, \quad y = \frac{D_2}{D} = \frac{-77}{11} = -7, \quad z = \frac{D_3}{D} = \frac{286}{11} = 26$$

Thus $x = -8, y = -7, z = 26$



Methods for solving Non-homogeneous System of Linear Equations

2. Matrix Inversion Method

Consider the system of n equations in n unknowns represented by

$$AX = B$$

Where A is square non-singular matrix.

Pre multiplying by A^{-1} on both sides, we get

$$A^{-1}AX = A^{-1}B$$

$$X = A^{-1}B$$

Which is the required solution.

Here A^{-1} is obtained by either by computing the adjoint or by use of elementary transformation.



Example

Solve by using Matrix Inversion Method

$$x_1 + x_2 + x_3 + x_4 = 0$$

$$x_1 + x_2 + x_3 - x_4 = 4$$

$$x_1 + x_2 - x_3 + x_4 = -4$$

$$x_1 - x_2 + x_3 + x_4 = 2$$

Solution: The system is written as $AX = B$ where

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \end{bmatrix}, X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 4 \\ -4 \\ 2 \end{bmatrix}$$

Computing inverse by elementary transformations

Consider $A = IA$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & -1 \\ 1 & 1 & -1 & 1 \\ 1 & -1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1, R_4 \rightarrow R_4 - R_1 \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & -2 & 0 \\ 0 & -2 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 1 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} A$$

$$R_2 \rightarrow \frac{-1}{2} R_2, R_3 \rightarrow \frac{-1}{2} R_3, R_4 \rightarrow \frac{-1}{2} R_4 \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/2 & -1/2 & 0 & 0 \\ 1/2 & 0 & -1/2 & 0 \\ 1/2 & 0 & 0 & -1/2 \end{bmatrix} A$$

$$R_2 \leftrightarrow R_4 \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1/2 & 0 & 0 & -1/2 \\ 1/2 & 0 & -1/2 & 0 \\ 1/2 & -1/2 & 0 & 0 \end{bmatrix} A$$

$$\begin{matrix} R_1 \rightarrow R_1 - R_2, R_1 \rightarrow R_1 - R_3, R_1 \rightarrow R_1 - R_4 \\ \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1/2 & 1/2 & 1/2 & 1/2 \\ 1/2 & 0 & 0 & -1/2 \\ 1/2 & 0 & -1/2 & 0 \\ 1/2 & -1/2 & 0 & 0 \end{bmatrix} A \end{matrix}$$

$$\text{Thus } A^{-1} = \begin{bmatrix} -1/2 & 1/2 & 1/2 & 1/2 \\ 1/2 & 0 & 0 & -1/2 \\ 1/2 & 0 & -1/2 & 0 \\ 1/2 & -1/2 & 0 & 0 \end{bmatrix}$$

Thus required solution is

$$X = A^{-1}B = \begin{bmatrix} -1/2 & 1/2 & 1/2 & 1/2 \\ 1/2 & 0 & 0 & -1/2 \\ 1/2 & 0 & -1/2 & 0 \\ 1/2 & -1/2 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 4 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \\ -2 \end{bmatrix}$$

$$\text{i.e., } x_1 = 1, x_2 = -1, x_3 = 2, x_4 = -2$$



Consistency of System of non-homogeneous Linear Equations

Consider system of m-linear equations in n-unknowns so that

$$AX = B$$

- Which can also be expressed as

$$x_1 \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix} + \dots + x_n \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

It is Clear that a solution exist if and only if the column matrix B can be expressed as a linear combination of columns of matrix A.



Consistency of System Non-Homogeneous of Linear Equations

We have seen that if a matrix is augmented by a row/column which is a linear combination of its rows/columns then the rank of augmented matrix equals rank of the matrix. Thus solution exists if and only if Rank of $[A|B]$ = Rank of A.

If Rank A = Rank $[A|B]$ then the system is consistent and

a. If $\text{Rank}(A) = \text{Rank}([A|B]) = n$ (number of unknowns)
then unique solution exists.

b. If $\text{Rank}(A) = \text{Rank}([A|B]) < n$ (no. of unknowns)

then system has infinitely many solutions in $(n-r)$ arbitrary constants.

If Rank of $[A|B] >$ Rank of A. Then system is inconsistent.



Consistency of System of Non-Homogeneous Linear Equations

Working Rule:

1. Determine $\text{Rank}(A)$ and $\text{Rank}([A|B])$
2. If $\text{Rank}(A) \neq \text{Rank}([A|B])$, System inconsistent, no solutions.
3. $\text{Rank}(A) = \text{Rank}([A|B]) = n$, then unique solution exists obtained by using backward substitution method.
4. $\text{Rank}(A) = \text{Rank}([A|B]) < n$, then rewrite $x_1, x_2 \dots x_r$ variables in terms of $(n-r)$ variables and solve by Gauss Jordan Elimination method.



Examples

1. Discuss the consistency of the following system of equations

$$2x + 3y + 4z = 11$$

$$x + 5y + 7z = 15$$

$$3x + 11y + 13z = 25. \text{ If found consistent, solve it.}$$

Solution: Consider Augmented matrix $[A|B] = \begin{bmatrix} 2 & 3 & 4 & | & 11 \\ 1 & 5 & 7 & | & 15 \\ 3 & 11 & 13 & | & 25 \end{bmatrix}$

$$R_2 \leftrightarrow R_1 \quad [A|B] \sim \begin{bmatrix} 1 & 5 & 7 & | & 15 \\ 2 & 3 & 4 & | & 11 \\ 3 & 11 & 13 & | & 25 \end{bmatrix}$$

$$R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 3R_1 \quad [A|B] \sim \begin{bmatrix} 1 & 5 & 7 & | & 15 \\ 0 & -7 & -10 & | & -19 \\ 0 & -4 & -8 & | & -20 \end{bmatrix}$$



Solution continued....

$$R_2 \rightarrow \frac{-1}{7} R_2, R_3 \rightarrow \frac{-1}{4} R_3 \quad [A|B] \sim \begin{bmatrix} 1 & 5 & 7 & | & 15 \\ 0 & 1 & 10/7 & | & 19/7 \\ 0 & 1 & 2 & | & 5 \end{bmatrix}$$

$$R_3 \rightarrow R_3 - R_2 \quad [A|B] \sim \begin{bmatrix} 1 & 5 & 7 & | & 15 \\ 0 & 1 & 10/7 & | & 19/7 \\ 0 & 0 & 4/7 & | & 16/7 \end{bmatrix}$$

Rank $A = \text{Rank}[A|B] = 3 = \text{number of unknowns}$.

Thus this system has unique solution.

Now by backward substitution we have

$$\frac{4}{7}z = \frac{16}{7}, y + \frac{10}{7}z = \frac{19}{7}, x + 5y + 7z = 15 \Rightarrow \mathbf{z = 4, y = -3, x = 2}$$



Example

Check for consistency and solve the system.

$$x - y + z = 2$$

$$3x - y + 2z = -6,$$

$$3x + y + z = -18$$

Solution: $[A|B] = \begin{bmatrix} 1 & -1 & 1 & : & 2 \\ 3 & -1 & 2 & : & -6 \\ 3 & 1 & 1 & : & -18 \end{bmatrix}$

• $R_2 \rightarrow R_2 - 3R_1, R_3 \rightarrow R_3 - 3R_1$ $[A|B] \sim \begin{bmatrix} 1 & -1 & 1 & : & 2 \\ 0 & 2 & -1 & : & -12 \\ 0 & 4 & -2 & : & -24 \end{bmatrix}$



Solution continued

$$R_3 \rightarrow R_3 - 2R_2 \quad [A|B] \sim \begin{bmatrix} 1 & -1 & 1 & \vdots & 2 \\ 0 & 2 & -1 & \vdots & -12 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 \rightarrow \frac{R_2}{2} \quad [A|B] \sim \begin{bmatrix} 1 & -1 & 1 & \vdots & 2 \\ 0 & 1 & -\frac{1}{2} & \vdots & -6 \\ 0 & 0 & 0 & \vdots & 0 \end{bmatrix}$$

Rank[A|B]=2 , Rank A =2

Since Rank[A|B]=Rank(A)<3 the system is consistent and has infinite number of solution.

The equivalent system is $x - y + z = 2 \quad \dots (1)$

$$y - \frac{1}{2}z = -6 \quad \dots (2)$$



Solution continued

Put $z = k$ (free variable)

From (2) $y = -6 + \frac{k}{2}$ in (1) solve for x

$$x = \frac{-8-k}{2}, y = -6 + \frac{k}{2}, z = k$$



Example

3) Check whether the following system of equations is consistent or not.

$$2x - y + z = 4,$$

$$3x - y + z = 6,$$

$$4x - y + 2z = 7,$$

$$-x + y - z = 9$$

Solution:

$$[A|B] = \begin{bmatrix} 2 & -1 & 1 & : & 4 \\ 3 & -1 & 1 & : & 6 \\ 4 & -1 & 2 & : & 7 \\ -1 & 1 & -1 & : & 9 \end{bmatrix}$$



Solution continued

- $R_1 \leftrightarrow -R_4$ $[A|B] \sim \begin{bmatrix} 1 & -1 & 1 & \vdots & -9 \\ 3 & -1 & 1 & \vdots & 6 \\ 4 & -1 & 2 & \vdots & 7 \\ 2 & -1 & 1 & \vdots & 4 \end{bmatrix}$
- $R_2 \rightarrow R_2 - 3R_1, R_3 \rightarrow R_3 - 4R_1, R_4 \rightarrow R_4 - 2R_1$
- $[A|B] \sim \begin{bmatrix} 1 & -1 & 1 & \vdots & -9 \\ 0 & 2 & -2 & \vdots & 33 \\ 0 & 3 & -2 & \vdots & 43 \\ 0 & 1 & -1 & \vdots & 22 \end{bmatrix}$



Solution continued

$$R_4 \rightarrow \frac{R_4}{-11}$$

$$[A|B] \sim \begin{bmatrix} 1 & -1 & 1 & \vdots & -9 \\ 0 & 1 & -1 & \vdots & 22 \\ 0 & 0 & 1 & \vdots & -23 \\ 0 & 0 & 0 & \vdots & 1 \end{bmatrix}$$

$$A \sim \begin{bmatrix} 1 & -1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus Rank A=3

$$\text{Rank } [A]=3, \quad \text{Rank } [A|B]=4$$

Since $\text{Rank } [A|B] \neq \text{Rank } [A]$, the system is inconsistent.



Example

For the value of μ and λ such that the following system

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

Has (i) No solution (ii) unique solution (iii) infinite solution.

$$\bullet \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ \mu \end{bmatrix} \quad AX=B$$



Example

- $[A|B] = \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 1 & 2 & 3 & : & 10 \\ 1 & 2 & \lambda & : & \mu \end{bmatrix}$

- $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$ $[A|B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 1 & \lambda - 1 & : & \mu - 6 \end{bmatrix}$

- $R_3 \rightarrow R_3 - R_2,$ $[A|B] \sim \begin{bmatrix} 1 & 1 & 1 & : & 6 \\ 0 & 1 & 2 & : & 4 \\ 0 & 0 & \lambda - 3 & : & \mu - 10 \end{bmatrix}$



Solution

- (i) System has no solution for $\text{Rank}[A] < \text{Rank}[A | B]$
i.e $\lambda = 3, \mu \neq 10$
- (ii) System has unique solution for $\text{Rank}[A | B] = \text{Rank } A = 3$
i.e $\lambda \neq 3, \mu \text{ any value.}$
- (iii) System has infinitely many solutions for $\text{Rank}[A | B] = \text{Rank } A < 3$
i.e $\lambda = 3, \mu = 10.$



Problems

1. Test the consistency of the following equation and solve them if possible

i) $3x + 3y + 3z = 1, x + 2y = 4, 10y + 3z = -2, 2x - 3z - z = 5$

ii) $x_1 - x_2 + x_3 - x_4 + x_5 = 1, 2x_1 - x_2 + 3x_3 + 4x_5 = 2, x_1 - x_2 + x_3 - x_4 + x_5 = 1,$
 $3x_1 - 2x_2 + 2x_3 + x_4 + x_5 = 1, x_1 + x_3 + 2x_4 + x_5 = 0$

iii) $4x - 5y + z = 2; 3x + y - 2z = 9; x + 4y + z = 5$

2. Find the value of k for which the following system of equation.

$3x_1 - 2x_2 + 2x_3 = 3, x_1 + kx_2 - 3x_3 = 0, 4x_1 + x_2 + 2x_3 = 7$ are consistent.

3. Find the value of λ for which the system of equations

$x + y + 4z = 1, x + 2y - 2z = 1, \lambda x + y + z = 1$ will have unique solution.



Problems

4. For what value of λ and μ the following system of equations.

$2x + 3y + 5z = 9, 7x + 3y - 2z = 8, 2x + 3y + \lambda z = \mu$ will have
i) unique solution ii) no solution iii) infinitely many solutions

5. Solve the following

$$7x + 16y - 7z = 4; 2x + 5y - 3z = 3; x + y + 2z = 4.$$

6. Find the value of a and b for which the system has (i) no solution (ii) unique solution (iii) infinitely many solutions.

$$2x + 3y + 5z = 9; 7x + 3y - 2z = 8; 2x + 3y + az = b$$

7. Solve the system $x + y + z = 6, 2x - 3y + 4z = 8, x - y + 2z = 5$ by
i) Cramer's Rule ii) Matrix Inversion iii) Gauss – Jordan.



Linear Dependence And Independence of Vectors

Vectors(columns or row)

$X_1, X_2 \dots X_n$ are said to be **linearly dependent** if

i) All the vectors are of the same order.

ii) There exists n scalars $\lambda_1, \lambda_2 \dots \lambda_n$ **not all zero** such that

$$\lambda_1 X_1 + \lambda_2 X_2 + \dots + \lambda_n X_n = 0$$

Otherwise they are called **linearly independent**.

i.e. if $\lambda_1 X_1 + \lambda_2 X_2 + \dots + \lambda_n X_n = 0$

then $\lambda_1 = \lambda_2 = \dots = \lambda_n = 0$



Examples

1. *Examine the following vectors for linear dependence and find the relation if it exists.*

$$X_1 = (1, 2, 4), X_2 = (2, -1, 3), X_3 = (0, 1, 2), X_4 = (-3, 7, 2)$$

Solution:

Consider the equation

$$\begin{aligned} X_1\lambda_1 + X_2\lambda_2 + X_3\lambda_3 + X_4\lambda_4 &= \mathbf{0} \\ \lambda_1(1, 2, 4) + \lambda_2(2, -1, 3) + \lambda_3(0, 1, 2) + \lambda_4(-3, 7, 2) &= \mathbf{0} \\ \lambda_1 + 2\lambda_2 + 0\lambda_3 - 3\lambda_4 &= 0 \\ 2\lambda_1 - \lambda_2 + \lambda_3 + 7\lambda_4 &= 0 \\ 4\lambda_1 + 3\lambda_2 + 2\lambda_3 + 2\lambda_4 &= 0 \end{aligned}$$

This is the homogeneous system

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 2 & -1 & 1 & 7 \\ 4 & 3 & 2 & 2 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 2 & -1 & 1 & 7 \\ 4 & 3 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & -5 & 2 & 14 \end{bmatrix} R_2 \rightarrow R_2 - 2R_1, R_4 \rightarrow R_4 - 4R_1$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & -5 & 2 & 14 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & 1 & 1 \end{bmatrix} R_3 \rightarrow R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \text{ which is can written in equation form as}$$

$$\lambda_1 + 2\lambda_2 + 0\lambda_3 - 3\lambda_4 = 0$$

$$-5\lambda_2 + \lambda_3 + 13\lambda_4 = 0$$

$$\lambda_3 + \lambda_4 = 0$$

Let $\lambda_4 = t$, from $\lambda_3 + \lambda_4 = 0$ it follows that $\lambda_3 = -t$
 $-5\lambda_2 - t + 13t = 0, \lambda_2 = \frac{12t}{5}$

$$\lambda_1 + \frac{24t}{5} - 3t = 0, \quad \lambda_1 = \frac{-9t}{5}$$

$$\lambda_1 = \frac{-9t}{5}, \lambda_2 = \frac{12t}{5}, \lambda_3 = -t, \lambda_4 = t$$

Hence the given vectors are linearly dependent.

Substituting the values of λ in 1st equation, we get

$$\frac{-9t}{5} X_1 + \frac{12t}{5} X_2 - tX_3 + tX_4 = 0$$

$$\therefore \text{for } t=5 \text{ we have } 9X_1 - 12X_2 + 5X_3 - 5X_4 = 0$$

2. Is the system of vectors $X_1 = (2,2,1)^t$, $X_2 = (1,3,1)^t$, $X_3 = (1,2,2)^t$ linearly dependent.

Solution:

$$\text{Here } X_1 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}, X_2 = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, X_3 = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$$

Consider the equation $\lambda_1 X_1 + \lambda_2 X_2 + \lambda_3 X_3 = 0$

$$\lambda_1 \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} + \lambda_3 \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = 0$$

$$2\lambda_1 + \lambda_2 + \lambda_3 = 0$$

$$2\lambda_1 + 3\lambda_2 + 2\lambda_3 = 0$$

$$2\lambda_1 + \lambda_2 + 2\lambda_3 = 0$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 & 1 \\ 2 & 3 & 2 \\ 1 & 1 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix} R_1 \leftrightarrow R_3$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 2 \\ 2 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & -1 & -3 \end{bmatrix} R_2 \rightarrow R_2 - 2R_1, R_3 \rightarrow R_3 - 2R_1$$

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & -1 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & -5 \end{bmatrix} R_3 \rightarrow R_3 + R_2$$

Writing the system of equation using the equivalent matrix we have

$$\begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -2 \\ 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\lambda_1 + \lambda_2 + \lambda_3 = 0$$

$$-\lambda_2 + \lambda_3 = 0$$

$$-5\lambda_3 = 0$$

$$\Rightarrow \lambda_3 = 0, \lambda_2 = 0, \lambda_1 = 0$$

Thus non-zero values of $\lambda_1, \lambda_2, \lambda_3$ **do not exist** which can satisfy

$\mathbf{X}_1\lambda_1 + \mathbf{X}_2\lambda_2 + \mathbf{X}_3\lambda_3 = 0$. Hence given vectors are not linearly dependent i.e., linearly independent.

Practice Problem:

1) Solve the system of equations $x + y - 3z + 2w = 0$

$$2x - y + 2z - 3w = 0$$

$$3x - 2y + z - 4w = 0$$

$$-4x + y - 3z + w = 0$$

2) Solve

$$x + y + z + w = 0$$

$$x + 3y + 2z + 4w = 0$$

$$2x + z - w = 0$$

3) Solve $x + 3y - 2z = 0$

$$2x - y + 4z = 0$$

$$x - 11y + 14z = 0$$

4) Find the value of k such that the system of equations $2x + 3y - 2z = 0$

$$3x - y + 3z = 0$$

$$7x + ky - z = 0 \text{ has non-trivial solutions.}$$

Find the solutions.

5) Find the value of k for which the following system of equations

$$4x + 2y - 5z = 0,$$

$$x + ky + 2z = 0,$$

$2x + y - z = 0$ has non-trivial solution.

6) Examine the following system of vectors for linear dependence. If dependent, find the relation between them.

a) $X_1 = (1, -1, 1), X_2 = (2, 1, 1), X_3 = (3, 0, 2)$

b) $X_1 = (1, 2, 3), X_2 = (2, -2, 6)$

c) $X_1 = (1, 1, 1, 3), X_2 = (1, 2, 3, 4), X_3 = (2, 3, 4, 7)$

d) $X_1 = (1, -1, 2, 0), X_2 = (2, 1, 1, 1), X_3 = (3, -1, 2, -1), X_4 = (3, 0, 3, 1)$